



The REECO Soldering Robot is an example in which two or more robots can form cooperative assemblies and, for example, can work with other equipment, loader, and unloader stations. The robot uses programmable logic controller (PLC) system to control all auxiliary equipment, such as transport, soldering generator, wire feeder, and tip cleaning station.

Temperature and wattage control. Temperature control is one of the most important factors in soldering systems to avoid faulty soldering. Wattage is a parameter to indicate how rapidly the heating element can raise the tip temperature.

An iron rated at 25 watts may take over a minute to bring the tip temperature up to 660°F (about 349 degrees centigrade) from room temperature and the one rated at 130 watts will get there in a couple of seconds. Moreover, when the hot tip transfers heat to the cooler PCB pad and component leads, it can become colder. Hence the heating element is used to maintain tip temperature. Lower wattage soldering irons cannot replenish the heat lost and may lead to defective soldering. So, it is advisable to select stations equipped with minimum 60 watts heating power.

In contrast to the popular belief,



QUICK ET9484E soldering robot



REECO Soldering Robot (Credit: www.renex.pl/en)



Ersa VERSAFLOW ONE

the high-wattage iron will not destroy delicate components. A temperature-controlled soldering iron is designed to switch off the heater when the tip reaches a preset temperature. A powerful heating system will only ensure quick thermal recovery during soldering. The soldering station with 130 watts of heating power will not be of much use if the tip is allowed to get cold before the heating element kicks in.

So, remember to choose stations with closed-loop temperature control. A digital closed-loop temperature control system comes with a heating element, temperature sensor, and also a microprocessor control unit. It is a common feature to prevent both temperature undershoot and overshoot while soldering.

The Hakko FX-951 is a relatively more modern station with direct heat soldering technology that not only allows it to react faster to thermal changes but also maintain tip temperature. The tip in such stations is a composite design available as a unified package with heating element and thermocouple to avoid

thermal drift.

In the automated lines, QUICK ET9484E Soldering Robot is a new launch that features a robust hot iron system with a superior closed-loop temperature sensing feedback. Two fine-toothed gears work to pull solder from the solder spool. It then perforates the solder wire, allowing better out gassing of flux core, resulting in less solder ball and flux splatter defects.

Ersa's VERSAFLOW ONE is another model with improved heating in the cross profile, which significantly reduces machine's power requirements.

Heat technology. Induction soldering is a process where two or more metals are joined together by melting and flowing a filler metal (solder) into the joint interface. It does not involve melting of the joints. This often gives a bit weaker joint compared to a brazed one but is a desirable result for many sensitive electronics or small components. Curie Heat Technology is where the iron tips respond to the thermal demands of each solder joint by adjusting the power instantaneously, thereby meeting the exact requirements of the substrate component and solder material.

One of the most critical aspects in soldering is choosing the right solder tip, which helps to achieve uniform heating of component leads and pads on the PCB. Some soldering solutions, such as Metcal's SmartHeat Technology, can be used with the necessary tips and cartridges to ensure proper soldering. The cartridge in here is made of a heater of proprietary design and has a non-heating copper core and an outer heating layer of magnetic alloy. The composition of the alloy predetermines the tip temperature, and it can self-regulate maintaining constant temperature ($\pm 1.1^{\circ}\text{C}$).

More axes. Traditional robotic sol-